

Antiferromagnetic Ising Model vs the Kitaev-Chain Qubit Hamiltonian

Seminar notes

May 31, 2026

1 Purpose

The Jordan-Wigner mapping turns the Kitaev chain into a spin-chain Hamiltonian. At first sight this can look similar to an Ising model, especially at the sweet spot $t = \Delta$, where only one two-qubit coupling remains. These notes clarify the relationship between a standard antiferromagnetic Ising model and the qubit Hamiltonian used in this project.

The main message is:

The Kitaev-chain qubit Hamiltonian can be written as an Ising-like spin model, but its physical interpretation is not the same as an ordinary antiferromagnet. The spin variables are a fermionic encoding, and the topological information appears most naturally in parity sectors and non-local strings.

2 Background: Ising Spins, Operators, and Eigenstates

The Ising model starts from a very simple local degree of freedom: each site has a two-state variable. In the classical model this variable is written as

$$S_j = \pm 1. \quad (1)$$

One may call $S_j = +1$ “spin up” and $S_j = -1$ “spin down”, but the important point is only that every site has two possible values.

In the quantum version, the two states at site j become a two-dimensional Hilbert space. We usually choose the Z basis

$$|\uparrow\rangle_j = |0\rangle_j, \quad |\downarrow\rangle_j = |1\rangle_j, \quad (2)$$

with

$$Z_j |\uparrow\rangle_j = + |\uparrow\rangle_j, \quad Z_j |\downarrow\rangle_j = - |\downarrow\rangle_j. \quad (3)$$

Thus the classical variable S_j is represented by the eigenvalue of the Pauli operator Z_j :

$$S_j \longleftrightarrow Z_j. \quad (4)$$

If we use physical spin- $\frac{1}{2}$ operators, they are related to Pauli matrices by

$$S_j^\alpha = \frac{\hbar}{2} \sigma_j^\alpha, \quad \alpha = x, y, z. \quad (5)$$

In quantum-information notation we usually set $\hbar = 1$ and write the Hamiltonian directly with Pauli matrices X, Y, Z . This absorbs factors of $1/2$ into the coupling constants. For example,

$$J_z S_j^z S_{j+1}^z = \frac{J_z \hbar^2}{4} Z_j Z_{j+1}. \quad (6)$$

So when these notes write $J Z_j Z_{j+1}$, J is the effective Pauli-basis coupling, not necessarily the microscopic spin-coupling constant before the $S^z = (\hbar/2)Z$ conversion. For a chain of L spins, the computational basis states are tensor products

$$|s_0 s_1 \cdots s_{L-1}\rangle = |s_0\rangle \otimes |s_1\rangle \otimes \cdots \otimes |s_{L-1}\rangle, \quad (7)$$

where each s_j is either $\uparrow$ or $\downarrow$.

2.1 Two-spin bond energies

The elementary Ising interaction is one bond:

$$H_{j,j+1} = J Z_j Z_{j+1}. \quad (8)$$

Because $Z_j Z_{j+1}$ is diagonal in the Z basis, the four two-spin basis states are already eigenstates:

$$Z_j Z_{j+1} |\uparrow\uparrow\rangle = + |\uparrow\uparrow\rangle, \quad (9)$$

$$Z_j Z_{j+1} |\downarrow\downarrow\rangle = + |\downarrow\downarrow\rangle, \quad (10)$$

$$Z_j Z_{j+1} |\uparrow\downarrow\rangle = - |\uparrow\downarrow\rangle, \quad (11)$$

$$Z_j Z_{j+1} |\downarrow\uparrow\rangle = - |\downarrow\uparrow\rangle. \quad (12)$$

Therefore the bond energies are

State	$Z_j Z_{j+1}$ eigenvalue	Bond energy
$ \uparrow\uparrow\rangle$	+1	$+J$
$ \downarrow\downarrow\rangle$	+1	$+J$
$ \uparrow\downarrow\rangle$	-1	$-J$
$ \downarrow\uparrow\rangle$	-1	$-J$

For an antiferromagnet, $J > 0$, so the lower-energy bond states are

$$|\uparrow\downarrow\rangle, \quad |\downarrow\uparrow\rangle. \quad (13)$$

For a ferromagnet, $J < 0$, so the lower-energy bond states are

$$|\uparrow\uparrow\rangle, \quad |\downarrow\downarrow\rangle. \quad (14)$$

This sign convention is why the antiferromagnetic Ising model is usually written with $J > 0$ in $J \sum Z_j Z_{j+1}$.

2.2 Many-spin eigenstates without a transverse field

For

$$H_0 = J \sum_{j=0}^{L-2} Z_j Z_{j+1}, \quad (15)$$

every computational-basis product state is an eigenstate, because all bond operators commute:

$$[Z_j Z_{j+1}, Z_k Z_{k+1}] = 0. \quad (16)$$

If a configuration has N_{same} aligned bonds and N_{opp} opposite bonds, its energy is

$$E = J(N_{\text{same}} - N_{\text{opp}}), \quad N_{\text{same}} + N_{\text{opp}} = L - 1. \quad (17)$$

For $J > 0$, minimizing the energy means maximizing N_{opp} . On an open chain this gives the two staggered ground states

$$|\uparrow\downarrow\uparrow\downarrow\cdots\rangle, \quad |\downarrow\uparrow\downarrow\uparrow\cdots\rangle, \quad (18)$$

with ground-state energy

$$E_0 = -J(L - 1). \quad (19)$$

A domain wall is a local defect where one bond fails to alternate; each such bad bond changes its contribution from $-J$ to $+J$, costing

$$\Delta E_{\text{bad bond}} = 2J. \quad (20)$$

2.3 What the transverse field changes

The transverse-field Ising model adds a term such as

$$-h \sum_j X_j. \quad (21)$$

This term is not diagonal in the Z basis, because

$$X_j |\uparrow\rangle_j = |\downarrow\rangle_j, \quad X_j |\downarrow\rangle_j = |\uparrow\rangle_j. \quad (22)$$

Therefore the product states that were exact eigenstates of H_0 are no longer exact eigenstates when $h \neq 0$. The transverse field flips spins and creates quantum superpositions of different classical configurations. The competition between J and h is the source of the quantum phase transition.

3 Reference Model: The Antiferromagnetic Ising Chain

A classical nearest-neighbor Ising chain is often written as

$$H_{\text{Ising}} = J \sum_{j=0}^{L-2} S_j S_{j+1}, \quad S_j = \pm 1. \quad (23)$$

For $J > 0$, neighboring spins prefer opposite signs:

$$S_j S_{j+1} = -1. \quad (24)$$

Thus the two classical ground-state patterns are the two staggered configurations

$$(+, -, +, -, \dots), \quad (-, +, -, +, \dots). \quad (25)$$

The order parameter is local after including the alternating sign:

$$m_{\text{stag}} = \frac{1}{L} \sum_{j=0}^{L-1} (-1)^j S_j. \quad (26)$$

The quantum transverse-field antiferromagnetic Ising chain is commonly written as

$$H_{\text{AFI}} = J \sum_{j=0}^{L-2} Z_j Z_{j+1} - h \sum_{j=0}^{L-1} X_j, \quad J > 0. \quad (27)$$

Here the $Z_j Z_{j+1}$ term favors staggered Z -basis order, while the transverse field X_j creates quantum fluctuations. The transition is between an ordered antiferromagnetic phase and a disordered paramagnetic phase. In this model, the natural diagnostic is still local or nearly local: staggered magnetization and spin-spin correlations in the ordering direction.

4 Our Qubit Hamiltonian

The Jordan-Wigner transform maps the open Kitaev chain to

$$H_K = -\frac{\mu}{2} \sum_{j=0}^{L-1} Z_j + \frac{t - \Delta}{2} \sum_{j=0}^{L-2} X_j X_{j+1} + \frac{t + \Delta}{2} \sum_{j=0}^{L-2} Y_j Y_{j+1}. \quad (28)$$

This is a transverse-field anisotropic XY chain, not a generic $Z_j Z_{j+1}$ Ising chain. The spin operators in Eq. (28) encode fermion occupation and fermion parity. In particular,

$$n_j = c_j^\dagger c_j = \frac{I - Z_j}{2}, \quad P = \prod_{j=0}^{L-1} Z_j. \quad (29)$$

The Z field is the chemical-potential term, not an external magnetic field acting on physical spins.

At the sweet spot $t = \Delta$, the XX coupling cancels:

$$H_K(t = \Delta) = -\frac{\mu}{2} \sum_j Z_j + t \sum_{j=0}^{L-2} Y_j Y_{j+1}. \quad (30)$$

At $\mu = 0$, this becomes a pure nearest-neighbor YY model:

$$H_K(t = \Delta, \mu = 0) = t \sum_{j=0}^{L-2} Y_j Y_{j+1}. \quad (31)$$

Since $t > 0$ in our conventions, each $Y_j Y_{j+1}$ bond energetically prefers opposite Y eigenvalues. In this narrow algebraic sense, the sweet-spot Hamiltonian is antiferromagnetic in the Y basis.

5 Why the Similarity Is Limited

5.1 The ordering direction is different

The standard antiferromagnetic Ising chain has an interaction $Z_j Z_{j+1}$ and therefore orders in the Z basis. Our sweet-spot Hamiltonian has $Y_j Y_{j+1}$ interactions. A single-qubit basis rotation can map Y to Z :

$$R_x\left(\frac{\pi}{2}\right) Y R_x^\dagger\left(\frac{\pi}{2}\right) = Z. \quad (32)$$

After such a rotation, Eq. (31) looks like an Ising coupling. However, this is a basis change inside the qubit encoding. It does not turn the Majorana system into an ordinary magnetic material.

5.2 The field term has a fermionic meaning

In the transverse-field Ising model, the field competes with spin order. In the Kitaev-chain qubit Hamiltonian, the Z term comes from the chemical potential:

$$-\mu c_j^\dagger c_j \mapsto -\frac{\mu}{2} Z_j \quad \text{up to a constant convention.} \quad (33)$$

Changing μ changes the fermion filling tendency and drives the system through the topological transition at

$$|\mu| = 2t. \quad (34)$$

Thus the same-looking transverse field has a different microscopic origin.

5.3 The diagnostic is not local staggered magnetization

An antiferromagnetic Ising phase can be detected by local staggered order:

$$\langle (-1)^j Z_j \rangle \neq 0 \quad \text{or} \quad \langle Z_i Z_j \rangle \sim (-1)^{i-j}. \quad (35)$$

The Kitaev chain, by contrast, is a topological superconducting phase. In a fixed-parity eigenstate, single-edge Majorana operators flip parity and have zero expectation value. The robust observable is a parity-preserving non-local string, for example

$$O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}. \quad (36)$$

For $L = 4$, this is

$$O_{\text{edge}} = X_0 Z_1 Z_2 X_3. \quad (37)$$

This is the central difference for the seminar: the Ising antiferromagnet is organized by conventional symmetry breaking, while the Kitaev phase is diagnosed through parity, edge-mode degeneracy, and non-local strings.

6 Comparison Table

Feature	Antiferromagnetic Ising chain	Kitaev-chain qubit Hamiltonian
Typical form	$J \sum Z_j Z_{j+1} - h \sum X_j$	$-\frac{\mu}{2} \sum Z_j + \frac{t-\Delta}{2} \sum XX + \frac{t+\Delta}{2} \sum YY$
Coupling sign	$J > 0$ favors opposite neighboring spins	$t > 0$ at $t = \Delta$ favors opposite Y eigenvalues
Order	staggered local spin order	topological edge correlations
Natural diagnostic	staggered magnetization	parity gap and non-local string order
Degrees of freedom	physical or effective spins	qubits encoding fermions via Jordan-Wigner
Phase transition	symmetry-breaking transition	topological transition at $ \mu = 2t$

7 Gauge and Sign Conventions

The word “antiferromagnetic” depends on both the coupling sign and the chosen spin basis. For a two-site Pauli coupling

$$H = J_\alpha \sum_j \sigma_j^\alpha \sigma_{j+1}^\alpha, \quad (38)$$

$J_\alpha > 0$ favors opposite eigenvalues of σ^α , while $J_\alpha < 0$ favors equal eigenvalues. In one dimension, some sign changes can also be moved around by staggered basis rotations. Therefore, calling Eq. (31) “antiferromagnetic in the Y basis” is a statement about the qubit Hamiltonian’s algebraic form, not the main physical content of the Kitaev phase.

8 Takeaway

The sweet-spot Kitaev-chain Hamiltonian resembles an antiferromagnetic Ising chain after choosing the appropriate spin basis:

$$H_K(t = \Delta, \mu = 0) = t \sum_j Y_j Y_{j+1}. \quad (39)$$

But the project should not present the physics as ordinary antiferromagnetism. The qubits encode fermions, Z measures local occupation parity, and the important phase information is not a local staggered magnetization. It is the topological structure: nearly degenerate parity sectors, Majorana edge modes, and non-local strings such as $X_0 Z_1 \cdots Z_{L-2} X_{L-1}$.